CCNA 200-301
Complete Lab Manual

100 Hands-On Labs for Packet Tracer & GNS3

From Beginner to Advanced

Created for CCNA Candidates
Based on Official Exam Blueprint

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Contents

Introduction

How to Use This Manual

This manual contains **100 structured labs** designed to prepare you for the CCNA 200-301 exam. Labs progress from basic configuration to complex troubleshooting scenarios.

Prerequisites

- Cisco Packet Tracer (v8.0+) or GNS3 with Cisco IOS images
- Basic understanding of networking concepts
- 2-3 months of dedicated practice (1-2 hours daily)

Lab Structure

Each lab includes:

- **Objective:** What you will accomplish
- **Topology:** Device layout (text-based diagram)
- **Configuration Steps:** Commands with explanations
- **Verification:** Show commands and expected output
- **Troubleshooting:** Common issues and solutions

Conventions

- `monospaced text` = Commands and output
- **Bold text** = Important concepts
- *Italic text* = Variables (replace with your values)

1 Beginner Level: Foundation Labs

Week 1-2: Basic Navigation and Configuration

1.1 Lab 1: Basic Network Topology Setup

Objective

Create a simple network with 2 PCs and 1 switch, then connect them properly.

Topology

```
[PC1] --- [Switch1] --- [PC2]
(Fa0)       (Fa0/1)      (Fa0/2)
```

Configuration Steps

1. Open Packet Tracer and add:
   - 1 x 2960 Switch
   - 2 x Generic PCs
2. Connect devices using Copper Straight-Through cables:
   - PC1 Fa0 → Switch Fa0/1
   - PC2 Fa0 → Switch Fa0/2
3. Verify link lights turn green (indicates Layer 1 connectivity)

Verification

```
! On switch:
Switch> show interfaces status
Port    Name    Status      Vlan    Duplex  Speed  Type
Fa0/1           connected   1       auto    auto   10/100BaseTX
Fa0/2           connected   1       auto    auto   10/100BaseTX
```

Troubleshooting

- If lights are amber/orange → Check cable type (use straight-through for PC to Switch)
- If lights are off → Verify both devices are powered on

1.2 Lab 2: Connecting PCs via Switch

Objective

Establish Layer 2 connectivity between two PCs through a single switch.

Topology

Same as Lab 1.

Configuration Steps

1. Configure IP addresses on PCs:
   - PC1: 192.168.1.1/24
   - PC2: 192.168.1.2/24
2. No switch configuration needed initially (default VLAN 1)

Verification

```
! On PC1:
C:\> ping 192.168.1.2
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

Expected Result

Ping success rate: 100%

1.3 Lab 3: Basic Router Configuration

Objective

Configure a router's hostname, interfaces, and verify connectivity.

Topology

```
[PC1] --- [Router1] --- [PC2]
 .1         .254        .254         .2
(192.168.1.0/24)    (192.168.2.0/24)
```

Configuration Steps

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.254 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 192.168.2.254 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# write memory
```

Verification

```
R1# show ip interface brief
Interface              IP-Address      OK? Method Status Protocol
GigabitEthernet0/0     192.168.1.254   YES manual up     up
GigabitEthernet0/1     192.168.2.254   YES manual up     up
```

1.4 Lab 4: Assigning IP Addresses to PCs

Objective

Learn different methods of assigning IP addresses to end devices.

Configuration Steps

1. **Static IP Assignment (Windows CLI):**

```
PC> ipconfig 192.168.1.10 255.255.255.0 192.168.1.1

```

2. **Static IP via GUI:**
   - Desktop → IP Configuration
   - Enter: IP Address, Subnet Mask, Default Gateway

3. **DHCP Assignment (after Lab 55):**

```
PC> ipconfig /dhcp

```

Verification

```
PC> ipconfig /all
FastEthernet0 Connection:
   IP Address: 192.168.1.10
   Subnet Mask: 255.255.255.0
   Default Gateway: 192.168.1.1
   DHCP Enabled: No
```

1.5 Lab 5: Configuring Hostnames on Devices

Objective

Properly name network devices for identification and management.

Configuration Steps

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
SW1(config)# exit
SW1#

Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# exit
R1#
```

Naming Conventions

- **Switches:** SW-[Location]-[Number] (e.g., SW-MDF-01)
- **Routers:** R-[Location]-[Number] (e.g., R-EDGE-01)
- **PCs:** PC-[VLAN]-[Number] (e.g., PC-10-05)

Verification

```
SW1# show running-config | include hostname
hostname SW1
```

1.6 Lab 6: Setting Up Console Access

Objective

Secure and configure console port access for out-of-band management.

Configuration Steps

```
SW1> enable
SW1# configure terminal
SW1(config)# line console 0
SW1(config-line)# password cisco123
SW1(config-line)# login
SW1(config-line)# exec-timeout 5 30
SW1(config-line)# logging synchronous
SW1(config-line)# end
SW1# write memory
```

Configuration Explanations

- `password cisco123` - Sets console password
- `login` - Enables password authentication
- `exec-timeout 5 30` - Times out after 5 minutes, 30 seconds
- `logging synchronous` - Prevents console messages from interrupting typing

Verification

```
! Exit and reconnect to console
SW1# exit

Press Enter
Password: cisco123
SW1>
```

1.7 Lab 7: Configuring Enable Password & Secret

Objective

Understand the difference between enable password (unencrypted) and enable secret (encrypted).

Configuration Steps

```
R1> enable
R1# configure terminal
R1(config)# enable password cisco
R1(config)# enable secret cisco123
R1(config)# exit
R1# exit

! Re-enter privileged mode
R1> enable
Password: cisco123
R1#
```

Verification - Viewing the Difference

```
R1# show running-config | include enable
enable secret 5 $1$Q8I5$LQYqZxKdV8kKo7HkPQkP10
enable password cisco
```

Key Points

- **enable secret** uses MD5 hashing (Type 5)
- **enable password** is stored in plain text (Type 0)
- If both configured, enable secret takes precedence
- **Always use enable secret**

1.8 Lab 8: Saving Configuration (Startup vs Running)

Objective

Understand the difference between running-config (RAM) and startup-config (NVRAM).

Key Concepts

- **Running-config:** Active configuration in RAM (lost on reboot)
- **Startup-config:** Saved configuration in NVRAM (loaded on boot)

Configuration Steps

```
! View both configurations
R1# show running-config
R1# show startup-config

! Save running to startup
R1# copy running-config startup-config
Destination filename [startup-config]? [Enter]
Building configuration...
[OK]

! Alternative methods
R1# write memory
R1# copy run start

! Erase startup config (for lab cleanup)
R1# erase startup-config
Erasing the nvram filesystem will remove all configuration files!
    Continue? [confirm]
[OK]
```

Verification

```
R1# show version | include image
System image file is "flash:c2900-universalk9-mz.SPA.157-3.M3.bin"
Configuration register is 0x2102
```

1.9 Lab 9: Static IP Configuration on PCs

Objective

Manually configure IP settings on end devices without DHCP.

Topology

```
[PC1]---[Switch]---[PC2]
.10/24             .20/24
```

Configuration Steps

1. **On PC1 (Windows Command Line):**

```
PC> ipconfig 192.168.1.10 255.255.255.0 192.168.1.1

```

2. **On PC2:**

```
PC> ipconfig 192.168.1.20 255.255.255.0 192.168.1.1

```

3. **Verify DNS setting (optional):**

```
PC> ipconfig /dns 8.8.8.8

```

Verification

```
PC1> ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

PC1> arp -a
Interface: 192.168.1.10 --- 0x3
  Internet Address      Physical Address      Type
  192.168.1.20          aaaa.bbbb.cccc        dynamic
```

1.10 Lab 10: Basic Ping Connectivity Test

Objective

Use ping to verify Layer 3 connectivity and troubleshoot issues.

Ping Command Options

```
! Basic ping
PC> ping 192.168.1.1

! Extended ping (on router)
R1# ping
Protocol [ip]:
Target IP address: 192.168.2.1
Repeat count [5]: 10
Datagram size [100]: 1500
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.1.254
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
```

Interpreting Results

!!!!! Success (0% loss)
..... No response (100% loss)
!.!.! Intermittent loss (congestion)
U.U.U Destination unreachable (routing issue)
Q.Q.Q Source quench (congestion)

Troubleshooting Matrix

Symptom	Probable Cause
"Request timed out"	Wrong IP, switch port down, ACL blocking
"Destination unreachable"	Default gateway missing, wrong subnet mask
"TTL expired"	Routing loop, TTL too low
"Reply from 192.168.1.254: Destination host unreachable"	No ARP entry, wrong VLAN

1.11 Lab 11: Using ‘show‘ Commands Basics

Objective

Master essential show commands for monitoring and troubleshooting.

Essential Show Commands

```
! Interface status
R1# show ip interface brief
R1# show interfaces
R1# show interfaces description

! Routing
R1# show ip route
R1# show ip route [network]
R1# show ip protocols

! System information
R1# show version
R1# show running-config
R1# show startup-config
R1# show clock

! Layer 2
SW1# show mac address-table
SW1# show vlan brief
SW1# show spanning-tree

! Security
R1# show ip ssh
R1# show users
R1# show privilege

! Filtering output
R1# show running-config | include hostname
R1# show running-config | section line
R1# show running-config | begin interface
R1# show ip interface brief | exclude unassigned
```

Output Filtering Examples

```
! Show only FastEthernet interfaces
R1# show ip interface brief | include FastEthernet

! Show lines 5-10
R1# show running-config | include FastEthernet

! Count occurrences
R1# show running-config | count interface
```

1.12 Lab 12: Switch MAC Address Table Exploration

Objective

Understand how switches learn MAC addresses and forward frames.

Topology

```
[PC1]---[SW1]---[PC2]
MAC A           MAC B
```

Configuration Steps

1. Connect PC1 and PC2 to SW1
2. Configure IP addresses on PCs
3. Generate traffic with ping

Verification

```
! View empty MAC table (before traffic)
SW1# show mac address-table
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

! Generate traffic from PC1 to PC2
PC1> ping 192.168.1.2

! View learned MAC addresses
SW1# show mac address-table
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    aaaa.bbbb.cccc    DYNAMIC     Fa0/1
   1    dddd.eeee.ffff    DYNAMIC     Fa0/2

! Clear MAC table
SW1# clear mac address-table dynamic

! Set static MAC entry
SW1# configure terminal
SW1(config)# mac address-table static aaaa.bbbb.cccc vlan 1 interface
    fa0/1
```

MAC Learning Process

1. PC1 sends frame to PC2 (dest MAC = PC2's MAC)
2. Switch records PC1's MAC and port (Fa0/1)
3. Switch floods frame out all ports (except Fa0/1)

4. PC2 responds, switch records PC2's MAC (Fa0/2)
5. Subsequent frames are forwarded directly (no flooding)

1.13 Lab 13: Configuring Duplex and Speed

Objective

Manually set interface speed and duplex to resolve negotiation issues.

Configuration Steps

```
SW1> enable
SW1# configure terminal
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# speed 100
SW1(config-if)# duplex full
SW1(config-if)# no shutdown

! Configure matching settings on connected device
! On PC1 or connected router, set same speed/duplex
```

Speed/Duplex Options

Speed	Values	Best Practice
10	10 Mbps	Legacy only
100	100 Mbps	FastEthernet
1000	1 Gbps	GigabitEthernet
auto	Auto-negotiate	**Default (recommended)**

Duplex Options

Type	Use Case	Description
Full-duplex	Switch to switch	Send/receive simultaneously
Half-duplex	Hub or old devices	One direction at a time
auto	Most modern links	Negotiate best mode

Troubleshooting

```
! Check for duplex mismatch
SW1# show interfaces fa0/1 | include duplex
Full-duplex, 100Mb/s, media type is 10/100BaseTX

! Look for collisions (half-duplex only)
SW1# show interfaces fa0/1 | include collisions
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
```

Verification

```
SW1# show interfaces status
Port    Name    Status     Vlan    Duplex  Speed  Type
Fa0/1           connected  1       full    100    10/100BaseTX
```

1.14 Lab 14: Introduction to CDP (Cisco Discovery Protocol)

Objective

Use CDP to discover neighboring Cisco devices automatically.

Topology

```
[R1] --- [SW1] --- [R2]
(g0/0)   (fa0/1)   (g0/1)
```

Configuration Steps

```
! CDP is enabled by default on Cisco devices
! View CDP neighbors
R1# show cdp neighbors
Capability Codes: R - Router, S - Switch, H - Host
Device ID        Local Intrfce     Holdtme    Capability  Platform
    Port ID
SW1              Gig 0/0           156        S I         WS-C2960   Fa0
    /1

! Detailed neighbor information
R1# show cdp neighbors detail

! View CDP traffic statistics
R1# show cdp traffic

! CDP timers (default: 60 sec hello, 180 sec hold)
R1# show cdp interface
```

CDP Configuration

```
! Disable CDP globally
R1(config)# no cdp run

! Re-enable CDP
R1(config)# cdp run

! Disable CDP on specific interface
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no cdp enable

! Change CDP timers
R1(config)# cdp timer 30
R1(config)# cdp holdtime 120
```

Verification

```
R1# show cdp neighbors
R1# show cdp entry *
```

1.15 Lab 15: Basic Switch Security (Port Shutdown)

Objective

Secure unused switch ports by administratively shutting them down.

Configuration Steps

```
SW1> enable
SW1# configure terminal

! Shutdown multiple unused ports
SW1(config)# interface range fastEthernet 0/10-24
SW1(config-if-range)# shutdown
SW1(config-if-range)# description ***UNUSED PORT - ADMIN DOWN***

! Verify
SW1(config-if-range)# end
SW1# show interfaces status | include down
Fa0/10          ***UNUSED P down    1        auto   auto  10/100BaseTX
```

Security Best Practices

1. Shut down all unused ports
2. Put ports in a "dead-end" VLAN (e.g., VLAN 999)
3. Add descriptive comments
4. Document in network diagram

Verification

```
! Check interface status
SW1# show interfaces fa0/10
FastEthernet0/10 is administratively down, line protocol is down

! Check specific port security
SW1# show interfaces description
Interface                      Status           Description
Fa0/10                         administratively down ***UNUSED PORT -
   ADMIN DOWN***
```

Audit Script

```
! Create audit of all unused ports
SW1# show run | include interface|shutdown|description
interface FastEthernet0/10
 shutdown
 description ***UNUSED PORT - ADMIN DOWN***
```

1.16 Lab 16: Configuring Banner MOTD

Objective

Configure legal notification banners for network device access.

Configuration Steps

```
R1> enable
R1# configure terminal

! Message of the Day banner (shown before login)
R1(config)# banner motd #
Enter TEXT message. End with the character '#'.
**********************************************
*      UNAUTHORIZED ACCESS PROHIBITED        *
*                                            *
* This system is for authorized users only.  *
* All activities are monitored and recorded. *
* By accessing this system, you consent      *
* to monitoring.                             *
**********************************************
#

! Login banner (shown after MOTD, before password prompt)
R1(config)# banner login #
Enter TEXT message. End with the character '#'.
Authorized Personnel Only
#

! Exec banner (shown after successful login)
R1(config)# banner exec #
Enter TEXT message. End with the character '#'.
Welcome to R1 - Production Router
#
```

Verification

```
! Exit and reconnect
R1# exit

**********************************************
*      UNAUTHORIZED ACCESS PROHIBITED        *
*                                            *
* This system is for authorized users only.  *
* All activities are monitored and recorded. *
* By accessing this system, you consent      *
* to monitoring.                             *
**********************************************

Authorized Personnel Only
User Access Verification
Password:
Welcome to R1 - Production Router
R1>
```

Legal Considerations

- Must specify authorized use only
- Should mention monitoring
- No misleading statements
- Clear ownership notice

1.17 Lab 17: Telnet Access to Router

Objective

Configure Telnet (unencrypted) for remote access (lab use only).

Topology

```
    [PC] --- [Switch] --- [R1]
     .10              .254 (vlan1)
     (VTY access)
```

Configuration Steps

```
R1> enable
R1# configure terminal

! Configure VTY lines (0-4 = 5 concurrent sessions)
R1(config)# line vty 0 4
R1(config-line)# password telnet123
R1(config-line)# login
R1(config-line)# transport input telnet
R1(config-line)# exec-timeout 10 0
R1(config-line)# logging synchronous
R1(config-line)# exit

! Configure interface for management
R1(config)# interface vlan 1
R1(config-if)# ip address 192.168.1.254 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit

! Enable Telnet server (default is enabled)
R1(config)# ip telnet server
```

Verification (from PC)

```
PC> telnet 192.168.1.254
Trying 192.168.1.254 ...Open
User Access Verification
Password: telnet123
R1>
```

Security Warning

- Telnet sends passwords in plain text
- **Use SSH in production**
- Only use Telnet in lab environments

1.18 Lab 18: SSH Configuration on Router

Objective

Configure secure SSH access for encrypted remote management.

Prerequisites

- Router must have a hostname and domain name
- Must have RSA keys generated
- Must have a username/password or AAA

Configuration Steps

```
R1> enable
R1# configure terminal

! Step 1: Set hostname and domain name
R1(config)# hostname R1
R1(config)# ip domain-name ccnadomain.com

! Step 2: Generate RSA keys (minimum 1024 bits for SSH v2)
R1(config)# crypto key generate rsa modulus 2048
The name for the keys will be: R1.ccnadomain.com
% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

! Step 3: Create local user
R1(config)# username admin secret cisco123

! Step 4: Configure VTY lines for SSH only
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exec-timeout 5 0
R1(config-line)# exit

! Step 5: Enable SSH version 2 (disable version 1)
R1(config)# ip ssh version 2

! Step 6: Optional - SSH timeouts
R1(config)# ip ssh time-out 60
R1(config)# ip ssh authentication-retries 3
```

Verification

```
! Check SSH status
R1# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3

! Check RSA keys
R1# show crypto key mypubkey rsa
```

```

! Test from PC
PC> ssh -l admin 192.168.1.254
Password: cisco123
R1>
```

1.19 Lab 19: Interface Status Verification

Objective

Master interface troubleshooting using various show commands.

Interface States Explained

Status	Protocol	Meaning
up	up	Working correctly
up	down	Layer 1 OK, Layer 2 problem (encapsulation mismatch)
down	down	Cable disconnected, switch port shutdown
administratively down	down	Interface manually shutdown

Verification Commands

```
! Quick overview
R1# show ip interface brief
Interface                  IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0         192.168.1.1     YES manual up      up
GigabitEthernet0/1         unassigned      YES unset  up      down
Serial0/0/0                10.0.0.1        YES manual up      up

! Detailed interface statistics
R1# show interfaces gigabitEthernet 0/0
GigabitEthernet0/0 is up, line protocol is up
  Hardware is CN Gigabit Ethernet, address is aaaa.bbbb.cccc
  Internet address is 192.168.1.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
  reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:00, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog, 0 multicast, 0 pause input
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier, 0 pause output
     0 output buffer failures, 0 output buffers swapped out

! Check interface counters
```

```
R1# clear counters
Clear "show interface" counters on all interfaces [confirm]
```

1.20 Lab 20: Basic Troubleshooting (Layer 1 & 2)

Objective

Systematically troubleshoot physical and data link layer issues.

Troubleshooting Methodology

1. Identify the problem
2. Establish theory of probable cause
3. Test the theory
4. Establish action plan
5. Implement solution
6. Verify full functionality
7. Document findings

Common Layer 1 Issues

Issue	Symptom	Solution
Wrong cable type	No link light	Use straight-through for PC-Switch, crossover for switch-switch
Distance exceeded	Intermittent connectivity	Max 100m for copper, use fiber for longer distances
Faulty cable	CRC errors	Replace cable
Bad port	Interface down/down	Move to different port
Power issues	Device unreachable	Check power supply

Common Layer 2 Issues

Issue	Symptom	Solution
Duplex mismatch	High collisions	Set both ends to auto or same manual setting
VLAN mismatch	No connectivity	Verify both ends in same VLAN
Trunk issues	Intermittent traffic	Check native VLAN matching
MAC flooding	Network slow	Implement port security
STP blockage	No traffic flow	Verify root bridge placement

Troubleshooting Commands

```
! Layer 1 checks
R1# show interfaces | include line protocol
R1# show interfaces status
R1# show logging

! Layer 2 checks
SW1# show vlan brief
SW1# show mac address-table
SW1# show spanning-tree
SW1# show interfaces trunk

! Cable testing (if supported)
SW1# test cable-diagnostics tdr interface fa0/1
SW1# show cable-diagnostics tdr interface fa0/1
```

Ping Troubleshooting Flowchart

```
Start: ping fails
    |
    v
Check IP configuration -> Wrong IP? -> Fix IP
    |
    v (correct)
Check default gateway -> Missing? -> Add gateway
    |
    v (present)
Check switch port status -> Down? -> Enable port
    |
    v (up)
Check VLAN assignment -> Wrong VLAN? -> Reassign
    |
    v (correct)
Check ACLs -> Blocking? -> Modify ACL
    |
    v (no ACL)
Check routing -> No route? -> Add route
```

1.21 Lab 21: Password Recovery Basics

Objective

Perform password recovery on Cisco routers and switches.

Important Note

This is for educational purposes only. Only perform on devices you own.

Router Password Recovery Process

1. **Power cycle the router**
2. **Press Ctrl+Break during boot** (within 60 seconds)

```
rommon 1 >

```

3. **Change configuration register to ignore startup config**

```
rommon 1 > confreg 0x2142
rommon 2 > reset

```

4. **After reboot, enter privileged mode (no password)**

```
Router> enable
Router#

```

5. **Load startup config into running config**

```
Router# copy startup-config running-config
Destination filename [running-config]?

```

6. **Change passwords**

```
Router# configure terminal
Router(config)# enable secret newpassword
Router(config)# line console 0
Router(config-line)# password newpassword
Router(config-line)# login
Router(config-line)# exit
Router(config)# line vty 0 4
Router(config-line)# password newpassword
Router(config-line)# login
Router(config-line)# exit

```

7. **Reset configuration register back to normal**

```
Router(config)# config-register 0x2102
Router(config)# end
Router# copy running-config startup-config
Router# reload

```

Configuration Register Values

Value	Meaning
0x2102	Normal (load startup config)
0x2142	Ignore startup config
0x2101	Boot into ROM monitor

Switch Password Recovery (Different Process)

1. Power cycle switch
2. Hold "Mode" button for 15 seconds
3. Enter switch: prompt

```
switch:

```

4. Initialize flash

```
switch: flash_init
switch: load_helper

```

5. Rename config file

```
switch: rename flash:config.text flash:config.old
switch: boot

```

6. Recover configuration

```
Switch# rename flash:config.old flash:config.text
Switch# copy flash:config.text running-config

```

1.22 Lab 22: Using Help and Command Shortcuts

Objective

Master Cisco CLI productivity features.

Context-Sensitive Help

```
! List all commands starting with "sh"
R1# sh?
show  shutdown

! List commands starting with "show ip"
R1# show ip?
  access-list  arp  bgp  cache  dhcp  eigrp  flow  http  interface
  nat  ospf  protocols  route  sla  ssh  traffic

! Show command syntax
R1# show ?
R1# show ip ?
R1# show ip route ?

! Partial keyword help
R1# show ip ro?
  route
```

Command Shortcuts

Shortcut	Full Command	Use
en	enable	Enter privileged mode
conf t	configure terminal	Global config mode
int	interface	Interface config mode
sh	show	Display information
sh ip int br	show ip interface brief	Quick interface status
wr	write memory	Save config

Command History

```
! Show command history
R1# show history

! Increase history buffer
R1# terminal history size 100

! Recall previous commands
Ctrl+P or Up Arrow  - Previous command
Ctrl+N or Down Arrow - Next command

! Search history
Ctrl+R - Reverse search (type part of command)
```

Editing Shortcuts

Keystroke	Action
Ctrl+A	Move to beginning of line
Ctrl+E	Move to end of line
Ctrl+F or Right Arrow	Forward one character
Ctrl+B or Left Arrow	Back one character
Ctrl+D	Delete character at cursor
Ctrl+K	Delete from cursor to end of line
Ctrl+U	Delete entire line
Ctrl+W	Delete previous word
Tab	Complete command/parameter
?	Context-sensitive help

Command Aliases

```
! Create custom shortcuts
R1(config)# alias exec sb show ip interface brief
R1(config)# alias exec sr show running-config
R1(config)# alias exec sc show clock

! Use alias
R1# sb
Interface              IP-Address      OK? Method Status Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up     up

! Remove alias
R1(config)# no alias exec sb
```

1.23 Lab 23: Configuring Clock on Router

Objective

Set system clock and timezone for accurate logging and timestamps.

Configuration Steps

```
R1> enable
R1# configure terminal

! Manual clock setting (24-hour format)
R1(config)# clock timezone EST -5
R1(config)# clock summer-time EDT recurring
R1(config)# exit
R1# clock set 14:30:00 15 Mar 2024

! Alternative: Set calendar (hardware clock)
R1# calendar set 14:30:00 15 Mar 2024

! Sync calendar to system clock
R1# clock read-calendar
```

Verification

```
! Show current time
R1# show clock
14:30:15.123 EST Thu Mar 15 2024

! Show with milliseconds
R1# show clock detail
14:30:15.123 EST Thu Mar 15 2024
Time source is user configuration

! Show calendar (hardware clock)
R1# show calendar
14:30:20 EST Thu Mar 15 2024
```

Enable Timestamps

```
R1(config)# service timestamps debug datetime msec localtime show-
    timezone
R1(config)# service timestamps log datetime msec localtime show-
    timezone

! Now log messages will show accurate time
R1# show logging
*Mar 15 14:30:15.123 EST: %SYS-5-CONFIG_I: Configured from console
```

NTP Integration (Preview of Lab 58)

```
! Use NTP for automatic time sync
R1(config)# ntp server 0.pool.ntp.org
R1# show ntp status
Clock is synchronized, stratum 2
```

1.24 Lab 24: Backup Configuration via TFTP

Objective

Backup device configurations to a TFTP server for disaster recovery.

Topology

```
[R1] --- [Switch] --- [TFTP Server]
 .1                (192.168.1.100)
```

Prerequisites

- TFTP server configured (use PC with TFTP software)
- IP connectivity between router and TFTP server

Configuration Steps

```
! Verify connectivity to TFTP server
R1# ping 192.168.1.100
!!!!!

! Backup running-config
R1# copy running-config tftp
Address or name of remote host []? 192.168.1.100
Destination filename [R1-confg]? R1-running-config-2024-03-15
!!
[OK - 1245 bytes]

! Backup startup-config
R1# copy startup-config tftp
Address or name of remote host []? 192.168.1.100
Destination filename [R1-confg]? R1-startup-config
!!
[OK - 1245 bytes]

! Backup entire flash (IOS image)
R1# copy flash: tftp
Source filename []? c2900-universalk9-mz.SPA.157-3.M3.bin
Address or name of remote host []? 192.168.1.100
Destination filename [c2900-universalk9-mz.SPA.157-3.M3.bin]?
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 45875200 bytes]
```

Verification on TFTP Server

```
! On PC (TFTP server)
TFTP Server Log:
Received R1-running-config-2024-03-15 (1245 bytes)
Received R1-startup-config (1245 bytes)
Received c2900-universalk9-mz.SPA.157-3.M3.bin (45875200 bytes)
```

Multiple Device Backup Script

```
! On management PC (Windows batch file)
@echo off
echo Backing up R1...
tftp -i 192.168.1.1 PUT running-config R1-config.txt
echo Backing up SW1...
tftp -i 192.168.1.2 PUT running-config SW1-config.txt
echo Backup complete!
```

1.25 Lab 25: Restore Configuration from TFTP

Objective

Restore device configuration from TFTP backup after failure.

Restore Process

```
! Method 1: Merge configuration (adds to existing)
R1# copy tftp running-config
Address or name of remote host []? 192.168.1.100
Source filename []? R1-running-config-2024-03-15
Destination filename [running-config]?
!!
[OK - 1245 bytes]

! Method 2: Replace entire configuration
! First, erase current config
R1# write erase
Erasing the nvram filesystem will remove all configuration files!
    Continue? [confirm]
[OK]
R1# reload

! After reboot, restore
R1# copy tftp startup-config
Address or name of remote host []? 192.168.1.100
Source filename []? R1-startup-config
Destination filename [startup-config]?
!!
[OK - 1245 bytes]

! Reload to load restored config
R1# reload
```

Restore IOS Image from TFTP

```
! If router is in ROMmon mode
rommon 1 > IP_ADDRESS=192.168.1.1
rommon 2 > IP_SUBNET_MASK=255.255.255.0
rommon 3 > DEFAULT_GATEWAY=192.168.1.254
rommon 4 > TFTP_SERVER=192.168.1.100
rommon 5 > TFTP_FILE=c2900-universalk9-mz.SPA.157-3.M3.bin
rommon 6 > tftpdnld

! Or from IOS
R1# copy tftp flash:
Address or name of remote host []? 192.168.1.100
Source filename []? c2900-universalk9-mz.SPA.157-3.M3.bin
Destination filename [c2900-universalk9-mz.SPA.157-3.M3.bin]?
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

Verification

```
! Verify restored configuration
R1# show running-config
R1# show version
```

2 Intermediate Level: VLANs, Routing & Services

Week 3-5: Switching Technologies

2.1 Lab 26: Creating VLANs

Objective

Create VLANs to segment Layer 2 broadcast domains.

Topology

```
[PC1-VLAN10]---[SW1]---[PC2-VLAN20]
```

Configuration Steps

```
SW1> enable
SW1# configure terminal

! Create VLAN 10
SW1(config)# vlan 10
SW1(config-vlan)# name Sales
SW1(config-vlan)# exit

! Create VLAN 20
SW1(config)# vlan 20
SW1(config-vlan)# name Engineering
SW1(config-vlan)# exit

! Create VLAN 99 (Management)
SW1(config)# vlan 99
SW1(config-vlan)# name Management
SW1(config-vlan)# exit

! Assign ports to VLANs
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# exit

SW1(config)# interface fastEthernet 0/2
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 20
SW1(config-if)# end

! Configure VLAN interface (SVI) for management
SW1(config)# interface vlan 99
SW1(config-if)# ip address 192.168.99.2 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
SW1(config)# ip default-gateway 192.168.99.1
```

Verification

```
! Show VLANs
SW1# show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- ---------
   -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5,
   Fa0/6
10   Sales                            active    Fa0/1
20   Engineering                      active    Fa0/2
99   Management                       active

! Show VLAN information
SW1# show vlan id 10
VLAN ID: 10
Name: Sales
State: active

! Test isolation (PC1 cannot ping PC2 even with same subnet)
PC1> ping 192.168.1.20
Request timed out.
```

2.2 Lab 27: Assigning Ports to VLANs

Objective

Learn multiple methods for assigning switch ports to VLANs.

Configuration Methods

```
! Method 1: Single port
SW1(config)# interface fastEthernet 0/10
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 30
SW1(config-if)# exit

! Method 2: Range of ports
SW1(config)# interface range fastEthernet 0/11-20
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 40
SW1(config-if-range)# exit

! Method 3: Interface range with mixed types
SW1(config)# interface range gigabitEthernet 0/1-2, fastEthernet 0/24
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 50

! Method 4: VLAN database mode (older IOS)
SW1# vlan database
SW1(vlan)# vlan 60 name Marketing
SW1(vlan)# apply
SW1(vlan)# exit
```

Verification

```
! Verify port assignments
SW1# show interfaces status
Port     Name     Status      Vlan     Duplex  Speed Type
Fa0/1             connected   10       full    100   10/100BaseTX
Fa0/2             connected   20       full    100   10/100BaseTX
Fa0/10            notconnect  30       auto    auto   10/100BaseTX
Fa0/11-20         notconnect  40       auto    auto   10/100BaseTX

! Detailed port VLAN info
SW1# show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (Sales)
Trunking Native Mode VLAN: 1 (default)
```

Best Practices

- Use descriptive VLAN names

- Document port assignments
- Use interface ranges for efficiency
- Keep management VLAN separate

2.3 Lab 28: VLAN Trunking (802.1Q)

Objective

Configure 802.1Q trunks to carry multiple VLANs between switches.

Topology

```
 [SW1]---(Trunk)---[SW2]
   |                 |
[PC1-VLAN10]      [PC3-VLAN10]
   |                 |
[PC2-VLAN20]      [PC4-VLAN20]
```

Configuration Steps

```
! On SW1
SW1> enable
SW1# configure terminal
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport trunk encapsulation dot1q
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk native vlan 99
SW1(config-if)# switchport trunk allowed vlan 10,20,99
SW1(config-if)# end

! On SW2 (same configuration)
SW2> enable
SW2# configure terminal
SW2(config)# interface gigabitEthernet 0/1
SW2(config-if)# switchport trunk encapsulation dot1q
SW2(config-if)# switchport mode trunk
SW2(config-if)# switchport trunk native vlan 99
SW2(config-if)# switchport trunk allowed vlan 10,20,99
SW2(config-if)# end
```

Verification

```
! Verify trunk status
SW1# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gi0/1       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gi0/1       10,20,99

Port        Vlans allowed and active in management domain
Gi0/1       10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/1       10,20,99

! Verify allowed VLANs
SW1# show interfaces gigabitEthernet 0/1 switchport
Trunking Native Mode VLAN: 99 (Management)
```

```
Trunking VLANs Enabled: 10,20,99
Pruning VLANs Enabled: 2-1001

! Test connectivity across trunk
PC1 (VLAN10) ping PC3 (VLAN10) -> Successful
PC1 (VLAN10) ping PC4 (VLAN20) -> Fails (different VLANs)
```

2.4 Lab 29: Native VLAN Configuration

Objective

Understand and configure the native VLAN for untagged traffic on trunks.

Key Concepts

- Native VLAN carries untagged traffic on trunk
- Default native VLAN is VLAN 1
- Both ends must match to prevent VLAN hopping

Configuration Steps

```
! Change native VLAN on SW1
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport trunk native vlan 999
SW1(config-if)# end

! Change native VLAN on SW2 (MUST match)
SW2(config)# interface gigabitEthernet 0/1
SW2(config-if)# switchport trunk native vlan 999
SW2(config-if)# end

! Verify native VLAN
SW1# show interfaces trunk
Port        Mode          Encapsulation  Status        Native vlan
Gi0/1       on            802.1q         trunking      999

! Create isolated VLAN for native (security best practice)
SW1(config)# vlan 999
SW1(config-vlan)# name NATIVE_UNUSED
SW1(config-vlan)# exit

! Remove VLAN 1 from trunk allowed list
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport trunk allowed vlan remove 1
```

Security Issue: VLAN Hopping

```
! Misconfiguration that allows attack
! SW1: native vlan 1, SW2: native vlan 999
! Attacker can send double-tagged 802.1Q frames

! Prevention:
! 1. Use unused VLAN as native
! 2. Disable DTP (Dynamic Trunking Protocol)
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport nonegotiate
! 3. Explicitly set trunk mode
SW1(config-if)# switchport mode trunk
```

2.5 Lab 30: Inter-VLAN Routing (Router-on-a-Stick)

Objective

Route between VLANs using a single router interface with subinterfaces.

Topology

```
[PC1-VLAN10]---[SW1]---(Trunk)---[R1]
[PC2-VLAN20]---[SW1]             (.1)
     192.168.10.0/24          192.168.20.0/24
```

Configuration Steps

```
! On SW1 - Trunk to router
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk native vlan 99
SW1(config-if)# switchport trunk allowed vlan 10,20,99

! On R1 - Subinterfaces
R1> enable
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# exit

R1(config)# interface gigabitEthernet 0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# exit

R1(config)# interface gigabitEthernet 0/0.99
R1(config-subif)# encapsulation dot1Q 99 native
R1(config-subif)# ip address 192.168.99.1 255.255.255.0
R1(config-subif)# exit

! Enable physical interface
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no shutdown
```

Verification

```
! On R1 - Verify subinterfaces
R1# show ip interface brief
GigabitEthernet0/0         unassigned      YES unset  up      up
GigabitEthernet0/0.10      192.168.10.1    YES manual up      up
GigabitEthernet0/0.20      192.168.20.1    YES manual up      up
GigabitEthernet0/0.99      192.168.99.1    YES manual up      up

! On PC1 (VLAN10)
PC1> ipconfig 192.168.10.10 255.255.255.0 192.168.10.1

! On PC2 (VLAN20)
```

```
PC2> ipconfig 192.168.20.20 255.255.255.0 192.168.20.1

! Test inter-VLAN routing
PC1> ping 192.168.20.20
Reply from 192.168.20.20: bytes=32 time=1ms TTL=127
```

2.6 Lab 31: VLAN Troubleshooting

Objective

Systematically troubleshoot common VLAN and trunking issues.

Common Issues and Solutions

Issue	Solution
VLAN mismatch	Verify access VLAN on both sides: `show interfaces switchport`
Trunk not forming	Check encapsulation: `switchport trunk encapsulation dot1q`
Native VLAN mismatch	Verify native VLAN matches: `show interfaces trunk`
Missing VLAN on trunk	Check allowed VLAN list: `switchport trunk allowed vlan`
Wrong IP subnet	Verify PCs have correct gateway for their VLAN
STP blocking	Check spanning-tree: `show spanning-tree vlan 10`

Troubleshooting Commands

```
! Step 1: Verify VLAN existence
SW1# show vlan brief

! Step 2: Verify port assignments
SW1# show interfaces status
SW1# show interfaces fa0/1 switchport

! Step 3: Verify trunk
SW1# show interfaces trunk
SW1# show interfaces gi0/1 switchport

! Step 4: Verify MAC addresses
SW1# show mac address-table | include 10

! Step 5: Verify router subinterfaces
R1# show ip route
R1# show interfaces | include line protocol|encapsulation

! Step 6: Verify PC configuration
PC> ipconfig
PC> arp -a

! Step 7: Check for VLAN pruning
SW1# show interfaces trunk | include Pruning

! Debug commands (use cautiously)
SW1# debug trunk
SW1# debug vlan packet
```

Troubleshooting Flowchart

```
PC cannot ping gateway
    |
    v
Check PC IP config -> Wrong? -> Fix
    |
    v (correct)
Check switchport VLAN -> Wrong? -> Fix
    |
    v (correct)
Check trunk between switch and router -> Down? -> Fix
    |
    v (up)
Check router subinterface config -> Missing/error? -> Fix
    |
    v (correct)
Check router routing table -> Route exists? -> Add
```

2.7 Lab 32: VTP Configuration (Server/Client)

Objective

Configure VLAN Trunking Protocol to manage VLANs across switches.

Topology

```
[SW1]---(Trunk)---[SW2]---(Trunk)---[SW3]
(VTP Server)      (VTP Client)      (VTP Client)
```

Configuration Steps

```
! On SW1 (VTP Server)
SW1(config)# vtp mode server
SW1(config)# vtp domain CCNA
SW1(config)# vtp password cisco123
SW1(config)# vtp version 2
SW1(config)# end
SW1# show vtp status

! On SW2 (VTP Client)
SW2(config)# vtp mode client
SW2(config)# vtp domain CCNA
SW2(config)# vtp password cisco123
SW2(config)# vtp version 2

! On SW3 (VTP Client)
SW3(config)# vtp mode client
SW3(config)# vtp domain CCNA
SW3(config)# vtp password cisco123
SW3(config)# vtp version 2

! Now create VLAN on Server (SW1)
SW1(config)# vlan 100
SW1(config-vlan)# name Sales
SW1(config-vlan)# exit
SW1(config)# vlan 200
SW1(config-vlan)# name Engineering
```

Verification

```
! On SW2 (should see VLANs automatically)
SW2# show vlan brief
100  Sales                              active
200  Engineering                        active

! VTP status
SW2# show vtp status
VTP Version                      : 2
Configuration Revision           : 5
Maximum VLANs supported locally  : 255
Number of existing VLANs         : 10
VTP Operating Mode               : Client
VTP Domain Name                  : CCNA
```

```
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Enabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x7A 0x3B 0x4C 0x1D
Configuration last modified by 192.168.1.1 at 3-15-24 14:30:00

! Verify VTP advertisements
SW1# show vtp counters
```

VTP Versions

Version	Features
1	Basic VLAN propagation
2	Token Ring support, consistency checks
3	Extended VLANs (1006-4094), private VLANs

2.8 Lab 33: VTP Pruning

Objective

Configure VTP pruning to reduce unnecessary broadcast traffic on trunks.

Configuration Steps

```
! Enable VTP pruning on VTP Server
SW1(config)# vtp pruning

! Verify pruning
SW1# show vtp status
VTP Pruning Mode                    : Enabled

! Check pruned VLANs on trunk
SW1# show interfaces trunk
Port         Vlans in spanning tree forwarding state and not pruned
Gi0/1        1,100,200

! Note: VLANs not present on downstream switches are pruned
```

Pruning Benefits

- Reduces broadcast traffic on trunks
- Improves bandwidth utilization
- Only sends VLAN traffic where needed

Manual Pruning (without VTP)

```
! Manually prune VLANs on trunk
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport trunk allowed vlan remove 300-400
SW1(config-if)# switchport trunk allowed vlan add 100-200
```

2.9 Lab 34: Private VLAN Basics

Objective

Configure Private VLANs for isolation within the same VLAN.

PVLAN Components

- **Primary VLAN**: Main VLAN
- **Secondary VLANs**: Community or Isolated
- **Promiscuous Port**: Gateway/router connection
- **Isolated Port**: Can only talk to promiscuous ports
- **Community Port**: Can talk within community + promiscuous

Configuration Steps

```
! Enable private VLAN feature
SW1(config)# vlan 100
SW1(config-vlan)# private-vlan primary
SW1(config-vlan)# exit

! Create secondary VLANs
SW1(config)# vlan 101
SW1(config-vlan)# private-vlan isolated
SW1(config-vlan)# exit

SW1(config)# vlan 102
SW1(config-vlan)# private-vlan community
SW1(config-vlan)# exit

! Associate secondary with primary
SW1(config)# vlan 100
SW1(config-vlan)# private-vlan association 101,102
SW1(config-vlan)# exit

! Configure ports
! Promiscuous port (to router)
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport mode private-vlan promiscuous
SW1(config-if)# switchport private-vlan mapping 100 101,102

! Isolated port
SW1(config)# interface fastEthernet 0/10
SW1(config-if)# switchport mode private-vlan host
SW1(config-if)# switchport private-vlan host-association 100 101

! Community port
SW1(config)# interface fastEthernet 0/20
SW1(config-if)# switchport mode private-vlan host
SW1(config-if)# switchport private-vlan host-association 100 102
```

Verification

```
SW1# show vlan private-vlan
Primary Secondary Type               Ports
100     101       isolated           Fa0/10
100     102       community          Fa0/20

SW1# show interfaces private-vlan mapping
Interface   Primary VLAN   Secondary VLAN   Type
Gi0/1       100            101,102         promiscuous
Fa0/10      100            101             host (isolated)
Fa0/20      100            102             host (community)
```

2.10 Lab 35: Voice VLAN Setup

Objective

Configure Voice VLAN for IP phones with QoS marking.

Topology

```
[IP Phone]---[SW1]---[Router]
   |                (Voice VLAN 150)
[PC]               (Data VLAN 10)
```

Configuration Steps

```
! Create VLANs
SW1(config)# vlan 10
SW1(config-vlan)# name DATA
SW1(config-vlan)# exit

SW1(config)# vlan 150
SW1(config-vlan)# name VOICE
SW1(config-vlan)# exit

! Configure port for voice and data
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# switchport voice vlan 150
SW1(config-if)# mls qos trust cos
SW1(config-if)# auto qos voip cisco-phone
SW1(config-if)# spanning-tree portfast
SW1(config-if)# end
```

Verification

```
! Show voice VLAN
SW1# show interfaces fa0/1 switchport | include Voice
Voice VLAN: 150 (VOICE)

! Show QoS
SW1# show mls qos interface fa0/1
FastEthernet0/1
trust state: trust cos
trust mode: trust cos
COS override: dis
default COS: 0

! CDP shows phone
SW1# show cdp neighbors fa0/1
Device ID: SEP123456789ABC
Interface: Fa0/1, Port ID: Port 1
Platform: Cisco IP Phone 7960
```

2.11 Lab 36: Spanning Tree Protocol (STP) Basics

Objective

Understand STP operation and view STP status.

Topology

```
         [SW1]
        /     \
    [SW2]      [SW3]
       \        /
         [SW4]
```

Configuration Steps (No config needed - STP default)

```
! View root bridge
SW1# show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.1234.5678
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.1234.5678
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- --------
    --------------------------------
Gi0/1            Desg FWD 4         128.1    P2p
Gi0/2            Desg FWD 4         128.2    P2p
Fa0/1            Altn BLK 19        128.3    P2p

! View port roles
SW1# show spanning-tree interface fastEthernet 0/1
Vlan                Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- --------
    --------------------------------
VLAN0001            Altn BLK 19        128.3    P2p
```

STP Port States

State	Description
Blocking	No BPDU received, no data
Listening	Receiving BPDUs, no data (15 sec)
Learning	Building MAC table, no data (15 sec)
Forwarding	Normal operation
Disabled	Administratively down

2.12 Lab 37: Rapid STP Configuration

Objective

Configure Rapid Spanning Tree Protocol (802.1w) for faster convergence.

Configuration Steps

```
! Enable RSTP globally
SW1(config)# spanning-tree mode rapid-pvst
SW2(config)# spanning-tree mode rapid-pvst
SW3(config)# spanning-tree mode rapid-pvst

! Verify
SW1# show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0001
Extended system ID is enabled
Portfast Default is disabled

! Convergence comparison
! STP: 30-50 seconds
! RSTP: 1-6 seconds

! Configure edge ports (Portfast equivalent)
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# spanning-tree portfast edge
SW1(config-if)# spanning-tree bpduguard enable
```

RSTP Port Roles

Role	Description
Root	Forwarding toward root bridge
Designated	Forwarding away from root
Alternate	Backup to root port (blocking)
Backup	Backup to designated port
Edge	Portfast enabled (host facing)

2.13 Lab 38: Root Bridge Election

Objective

Control root bridge placement for optimal traffic flow.

Configuration Steps

```
! Method 1: Set priority (lower value wins)
SW1(config)# spanning-tree vlan 1 priority 4096

! Method 2: Use root primary macro
SW1(config)# spanning-tree vlan 1 root primary
! Sets priority to 24576 (or 4096 less than current root)

! Method 3: Use root secondary macro
SW2(config)# spanning-tree vlan 1 root secondary
! Sets priority to 28672

! Verify root bridge
SW1# show spanning-tree vlan 1 | include "This bridge is the root"
This bridge is the root

! Set priority for specific VLANs
SW1(config)# spanning-tree vlan 10 priority 4096
SW1(config)# spanning-tree vlan 20 priority 8192
SW1(config)# spanning-tree vlan 30 priority 12288

! Verify per-VLAN root
SW1# show spanning-tree vlan 10
SW1# show spanning-tree vlan 20
```

Priority Values

Priority	Usage
0	Highest (never used - conflicts with bridge ID)
4096	Primary root (use `root primary`)
8192	Secondary root
12288	Tertiary
32768	Default (all switches)
61440	Lowest possible

Best Practices

- Set root bridge on distribution/core switches
- Set secondary root on backup distribution switch
- Use per-VLAN root for load balancing
- Never use priority 0

2.14 Lab 39: PortFast & BPDU Guard

Objective

Configure PortFast for host ports and BPDU Guard for security.

Configuration Steps

```
! Configure PortFast on access ports
SW1(config)# interface range fastEthernet 0/1-24
SW1(config-if-range)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
    single host.
 SW1(config-if-range)# spanning-tree bpduguard enable

! Alternative: Global PortFast for all access ports
SW1(config)# spanning-tree portfast default
SW1(config)# spanning-tree portfast bpduguard default

! Verify PortFast
SW1# show spanning-tree interface fa0/1 detail
Port 3 (FastEthernet0/1) of VLAN0001 is forwarding
Port path cost 19, Port priority 128, Port Identifier 128.3.
Designated root has priority 32769, address 0001.1234.5678
Designated bridge has priority 32769, address 0001.1234.5678
Designated port id is 128.3, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is point-to-point by default
**PortFast is configured (portfast configured)**

! BPDU Guard action (if BPDU received)
SW1# show spanning-tree summary
Portfast BPDU Guard is enabled globally
```

PortFast Benefits

- Bypasses listening/learning states
- Immediate transition to forwarding
- Saves 30 seconds per port
- Safe only on host-facing ports

BPDU Guard Behavior

- Shuts down port if BPDU received
- Prevents rogue switch connections
- Port enters err-disabled state

Recovering from err-disabled

```
! Manual recovery
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# shutdown
SW1(config-if)# no shutdown

! Automatic recovery
SW1(config)# errdisable recovery cause bpduguard
SW1(config)# errdisable recovery interval 300
```

2.15 Lab 40: EtherChannel (LACP)

Objective

Configure Link Aggregation Control Protocol for bandwidth and redundancy.

Topology

```
[SW1]===========[SW2]
(2 or 4 port bundle)
```

Configuration Steps

```
! On SW1
SW1(config)# interface range gigabitEthernet 0/1-2
SW1(config-if-range)# channel-group 1 mode active
SW1(config-if-range)# exit
SW1(config)# interface port-channel 1
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk allowed vlan 10,20,30

! On SW2
SW2(config)# interface range gigabitEthernet 0/1-2
SW2(config-if-range)# channel-group 1 mode active
SW2(config-if-range)# exit
SW2(config)# interface port-channel 1
SW2(config-if)# switchport mode trunk
SW2(config-if)# switchport trunk allowed vlan 10,20,30
```

LACP Modes

Mode	Description
Active	Actively negotiates (sends LACP packets)
Passive	Responds to negotiations (doesn't initiate)
On	Static configuration (no negotiation)

Verification

```
! Show EtherChannel summary
SW1# show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)          LACP      Gi0/1(P)    Gi0/2(P)

```

```
! Show port-channel details
SW1# show etherchannel 1 detail
! Show load balancing
SW1# show etherchannel load-balance
EtherChannel Load-Balancing Configuration:
    src-mac

! Configure load balancing
SW1(config)# port-channel load-balance src-dst-ip
```

2.16 Lab 41: EtherChannel (PAgP)

Objective

Configure Port Aggregation Protocol (Cisco proprietary).

Configuration Steps

```
! On SW1
SW1(config)# interface range gigabitEthernet 0/1-2
SW1(config-if-range)# channel-group 1 mode desirable
SW1(config-if-range)# exit

! On SW2
SW2(config)# interface range gigabitEthernet 0/1-2
SW2(config-if-range)# channel-group 1 mode desirable
SW2(config-if-range)# exit

! Verify
SW1# show etherchannel summary
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)        PAgP       Gi0/1(P)    Gi0/2(P)
```

PAgP Modes

Mode	Description
Desirable	Actively negotiates
Auto	Responds passively

Comparison: LACP vs PAgP

Feature	LACP	PAgP
Standard	IEEE 802.3ad	Cisco proprietary
Multi-vendor	Yes	No
Maximum ports	16 (8 active)	8
Mode options	Active/Passive	Desirable/Auto

2.17 Lab 42: Static Routing

Objective

Configure static routes to reach remote networks.

Topology

```
[PC1]---[R1]---(10.0.0.0/30)---[R2]---[PC2]
 .1/24     .1/30            .2/30    .1/24
(192.168.1.0/24)                     (192.168.2.0/24)
```

Configuration Steps

```
! On R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown

R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown

! Static route to reach R2's LAN
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

! On R2
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown

R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown

! Static route back to R1's LAN
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification

```
! Check routing table
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

```

```
Gateway of last resort is not set

S     192.168.2.0/24 [1/0] via 10.0.0.2
C     192.168.1.0/24 is directly connected, GigabitEthernet0/0
L     192.168.1.1/32 is directly connected, GigabitEthernet0/0
C     10.0.0.0/30 is directly connected, Serial0/0/0
L     10.0.0.1/32 is directly connected, Serial0/0/0

! Test connectivity
PC1> ping 192.168.2.2
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
```

2.18 Lab 43: Default Route Configuration

Objective

Configure default routes for internet-bound traffic.

Topology

```
[LAN]---[R1]---(ISP)---[Internet]
         (.1)    (default route)
```

Configuration Steps

```
! On R1
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1

! Alternative: Using exit interface
R1(config)# ip route 0.0.0.0 0.0.0.0 serial 0/0/0

! Using both next-hop and interface
R1(config)# ip route 0.0.0.0 0.0.0.0 serial 0/0/0 203.0.113.1

! Verify
R1# show ip route
Gateway of last resort is 203.0.113.1 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 203.0.113.1

! Test
R1# ping 8.8.8.8
!!!!!
```

Default Route Use Cases

- Internet connectivity
- Stub networks (single exit point)
- Simplifying routing tables
- Combined with dynamic routing

2.19 Lab 44: Floating Static Routes

Objective

Configure backup routes with higher administrative distance.

Topology

```
[R1]---(Primary: Serial0/0/0, AD=1)---[R2]
 |                                     |
 |---(Backup: Serial0/0/1, AD=5)--------|
```

Configuration Steps

```
! Primary route (AD default is 1)
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

! Backup route (AD 5 - higher than OSPF's 110)
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.1.2 5

! Verify primary route
R1# show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
  Known via "static", distance 1, metric 0
  * 10.0.0.2

! Simulate primary link failure
R1(config)# interface serial 0/0/0
R1(config-if)# shutdown

! Check backup route now active
R1# show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
  Known via "static", distance 5, metric 0
  * 10.0.1.2

! Restore primary
R1(config-if)# no shutdown
! Route reverts to primary
```

Administrative Distance Values

Routing Protocol	AD
Connected interface	0
Static route	1
EIGRP summary	5
External BGP	20
EIGRP internal	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200
Unknown	255

2.20 Lab 45: RIP v2 Configuration

Objective

Configure RIPv2 for small network routing.

Topology

```
[R1]---(10.0.0.0/30)---[R2]---(10.0.1.0/30)---[R3]
(192.168.1.0/24)                    (192.168.3.0/24)
       (192.168.2.0/24 on R2 loopback)
```

Configuration Steps

```
! On R1
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# no auto-summary
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0
R1(config-router)# exit

! On R2
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 10.0.0.0
R2(config-router)# network 10.0.1.0
R2(config-router)# network 192.168.2.0
R2(config-router)# exit

! On R3
R3(config)# router rip
R3(config-router)# version 2
R3(config-router)# no auto-summary
R3(config-router)# network 192.168.3.0
R3(config-router)# network 10.0.1.0
```

Verification

```
! Check RIP database
R1# show ip rip database
192.168.1.0/24    auto-summary
192.168.1.0/24    directly connected, GigabitEthernet0/0
192.168.2.0/24    auto-summary
192.168.2.0/24
    [1] via 10.0.0.2, 00:00:25, Serial0/0/0
192.168.3.0/24    auto-summary
192.168.3.0/24
    [2] via 10.0.0.2, 00:00:25, Serial0/0/0

! Check RIP neighbors
R1# show ip rip neighbors
Neighbor         V    Version    Interface    Address
10.0.0.2         2    2          Serial0/0/0  10.0.0.2
```

```

! Debug RIP (use cautiously)
R1# debug ip rip
RIP protocol debugging is on
R1#
*Mar 15 14:30:15.123: RIP: sending v2 update to 224.0.0.9 via Serial0
   /0/0
*Mar 15 14:30:15.123: RIP: build update entries
*Mar 15 14:30:15.123:    192.168.1.0/24 via 0.0.0.0, metric 1, tag 0
```

2.21 Lab 46: RIP Troubleshooting

Objective

Troubleshoot common RIP configuration issues.

Common Issues

Issue	Solution
No routes exchanged	Verify `no auto-summary`
RIPv1 vs RIPv2 mismatch	Ensure both use `version 2`
Split horizon issues	Check `no ip split-horizon` on frame relay
Route flapping	Increase timers: `timer basic 30 90 120 180`
Passive interfaces	Check `passive-interface default`
Authentication fails	Configure key chain on both sides

Troubleshooting Commands

```
! Verify RIP is running
R1# show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 18 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface        Send  Recv  Triggered RIP  Key-chain
    GigabitEthernet0/0 2     2
    Serial0/0/0       2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    192.168.1.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.2             120      00:00:03
  Distance: (default is 120)

! Check interface for RIP
R1# show ip rip interface
Serial0/0/0    (RIP version 2)
    Sending    (updates every 30 sec)
    Next due in 12 seconds
    Invalid after 180 sec
    Hold down 180 sec
    Flushed after 240 sec
```

2.22 Lab 47: Single-Area OSPF

Objective

Configure OSPF in a single area (Area 0).

Topology

```
[R1]---(10.0.0.0/30)---[R2]---(10.0.1.0/30)---[R3]
(.1/24)               (.2/.1)               (.2/24)
(Lo0:1.1.1.1/32)      (Lo0:2.2.2.2/32)      (Lo0:3.3.3.3/32)
```

Configuration Steps

```
! On R1
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# exit

! On R2
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 10.0.1.0 0.0.0.3 area 0
R2(config-router)# network 2.2.2.2 0.0.0.0 area 0
R2(config-router)# exit

! On R3
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 10.0.1.0 0.0.0.3 area 0
R3(config-router)# network 3.3.3.3 0.0.0.0 area 0
```

Verification

```
! Check OSPF neighbors
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    10.0.0.2        Serial0/0/0

! Check OSPF routes
R1# show ip route ospf
O        192.168.3.0/24 [110/65] via 10.0.0.2, 00:00:10, Serial0/0/0
O        2.2.2.2/32 [110/65] via 10.0.0.2, 00:00:10, Serial0/0/0
O        3.3.3.3/32 [110/129] via 10.0.0.2, 00:00:10, Serial0/0/0

! Check OSPF database
R1# show ip ospf database
```

```

! Test connectivity
R1# ping 192.168.3.1 source 192.168.1.1
!!!!!
```

2.23 Lab 48: OSPF Cost Manipulation

Objective

Change OSPF path selection by modifying interface costs.

Configuration Steps

```
! Method 1: Set cost directly on interface
R1(config)# interface serial 0/0/0
R1(config-if)# ip ospf cost 100

! Method 2: Change reference bandwidth
R1(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.

! Method 3: Change interface bandwidth (affects cost)
R1(config-if)# bandwidth 1000
! Cost = Reference_BW / Bandwidth (default ref=100Mbps)

! Verify costs
R1# show ip ospf interface serial 0/0/0
Serial0/0/0 is up, line protocol is up
  Internet Address 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost: 100

! See entire OSPF topology
R1# show ip ospf topology

! OSPF path selection order:
! 1. Intra-area routes (O)
! 2. Inter-area routes (O IA)
! 3. E1 external routes
! 4. E2 external routes
```

Reference Bandwidth Best Practices

```
! For modern networks with 10Gbps+
R1(config-router)# auto-cost reference-bandwidth 100000
! This sets 10Gbps to cost 1
```

2.24 Lab 49: OSPF Router ID Configuration

Objective

Configure and manage OSPF Router IDs.

Configuration Steps

```
! Method 1: Manual Router ID
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1

! Method 2: Loopback interface (highest IP)
R1(config)# interface loopback 0
R1(config-if)# ip address 10.255.255.1 255.255.255.255

! Method 3: Highest active interface IP (if no loopback)

! Router ID selection order:
! 1. Manual router-id command
! 2. Highest loopback IP address
! 3. Highest active interface IP address

! Change Router ID (requires OSPF restart)
R1# clear ip ospf process
Reset ALL OSPF processes? [no]: yes

! Verify Router ID
R1# show ip ospf
 Routing Process "ospf 1" with ID 1.1.1.1
 Start time: 00:00:15.123, Time elapsed: 00:10:00.456
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
 Supports area transit capability

! Check neighbor Router IDs
R1# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    10.0.0.2        Serial0/0/0
```

2.25 Lab 50: Passive Interfaces in OSPF

Objective

Configure passive interfaces to stop OSPF hellos on LAN segments.

Configuration Steps

```
! Make LAN interface passive (stops OSPF advertisements but still
  includes network)
R1(config)# router ospf 1
R1(config-router)# passive-interface gigabitEthernet 0/0

! Make all interfaces passive, then enable specific ones
R1(config-router)# passive-interface default
R1(config-router)# no passive-interface serial 0/0/0

! Verify passive interfaces
R1# show ip ospf interface
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  **Passive interface, No hellos**

Serial0/0/0 is up, line protocol is up
  Internet Address 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost:
  64
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

! Check OSPF neighbors (only active interfaces)
R1# show ip ospf neighbor
```

Benefits of Passive Interfaces

- Reduces OSPF overhead
- Improves security (no OSPF adjacencies on LAN)
- Still advertises LAN subnet
- Best practice for end-user segments

2.26 Lab 51: OSPF Troubleshooting

Objective

Troubleshoot common OSPF neighbor and route issues.

Common OSPF Issues

Issue	Solution
Neighbors stuck in INIT	Verify multicast reachability (224.0.0.5/6)
Neighbors stuck in 2-WAY	Check DR/BDR election (need Full on broadcast)
Neighbors stuck in EXSTART/EXCHANGE	MTU mismatch, check MTU on interfaces
No routes in table	Verify network statements, area ID matching
Authentication failure	Configure same key/password on both ends
Area mismatch	Check area ID (must match)

Troubleshooting Commands

```
! Check neighbor state
R1# show ip ospf neighbor
R1# show ip ospf neighbor detail

! Check OSPF interface settings
R1# show ip ospf interface
R1# show ip ospf interface serial 0/0/0

! Debug OSPF (use cautiously)
R1# debug ip ospf adj
R1# debug ip ospf events
R1# debug ip ospf packet

! Common MTU mismatch debug output
*Mar 15 14:30:15.123: OSPF-1 ADJ Ser0/0/0: Send DBD to 10.0.0.2
*Mar 15 14:30:15.124: OSPF-1 ADJ Ser0/0/0: Rcv DBD from 10.0.0.2
*Mar 15 14:30:15.124: OSPF-1 ADJ Ser0/0/0: Nbr 2.2.2.2 has larger MTU
! Fix with:
R1(config-if)# mtu 1500

! Clear OSPF process
R1# clear ip ospf process

! Check OSPF database for inconsistencies
R1# show ip ospf database
R1# show ip ospf database router
R1# show ip ospf database network
```

2.27 Lab 52: EIGRP Basic Configuration

Objective

Configure EIGRP for basic routing.

Configuration Steps

```
! On R1
R1(config)# router eigrp 100
R1(config-router)# network 192.168.1.0 0.0.0.255
R1(config-router)# network 10.0.0.0 0.0.0.3
R1(config-router)# network 1.1.1.1 0.0.0.0
R1(config-router)# no auto-summary
R1(config-router)# exit

! On R2
R2(config)# router eigrp 100
R2(config-router)# network 10.0.0.0 0.0.0.3
R2(config-router)# network 10.0.1.0 0.0.0.3
R2(config-router)# network 2.2.2.2 0.0.0.0
R2(config-router)# no auto-summary

! On R3
R3(config)# router eigrp 100
R3(config-router)# network 192.168.3.0 0.0.0.255
R3(config-router)# network 10.0.1.0 0.0.0.3
R3(config-router)# network 3.3.3.3 0.0.0.0
R3(config-router)# no auto-summary
```

Verification

```
! Check EIGRP neighbors
R1# show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address         Interface        Hold Uptime   SRTT   RTO  Q   Seq
                                     (sec)         (ms)        Cnt Num
0   10.0.0.2        Se0/0/0          11 00:00:15  1      100   0   3

! Check EIGRP topology
R1# show ip eigrp topology
EIGRP-IPv4 Topology Table for AS(100)/ID(1.1.1.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 192.168.1.0/24, 1 successors, FD is 28160
        via Connected, GigabitEthernet0/0
P 192.168.3.0/24, 1 successors, FD is 2172416
        via 10.0.0.2 (2172416/2169856), Serial0/0/0

! Check EIGRP routes
R1# show ip route eigrp
D    192.168.3.0/24 [90/2172416] via 10.0.0.2, 00:00:10, Serial0/0/0
```

2.28 Lab 53: EIGRP Metrics and Feasible Distance

Objective

Understand EIGRP metric calculation and successor selection.

Configuration Steps

```
! View EIGRP topology with metrics
R1# show ip eigrp topology all-links
EIGRP-IPv4 Topology Table for AS(100)/ID(1.1.1.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 192.168.3.0/24, 1 successors, FD is 2172416, serno 3
        via 10.0.0.2 (2172416/2169856), Serial0/0/0
        via 10.0.1.2 (2172416/2169856), Serial0/0/1

! Modify EIGRP metric weights
R1(config-router)# metric weights 0 1 0 1 0 0
! K1=Bandwidth, K2=Load, K3=Delay, K4=Reliability, K5=MTU

! Default formula: Metric = (Bandwidth + Delay) * 256
! Bandwidth = 10^7 / min BW in kbps
! Delay = sum of delays in 10s of microseconds

! Change interface bandwidth/delay
R1(config)# interface serial 0/0/0
R1(config-if)# bandwidth 1544
R1(config-if)# delay 20000

! Verify metric changes
R1# show ip eigrp topology 192.168.3.0/24
EIGRP-IPv4 (AS 100): Topology entry for 192.168.3.0/24
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is
   40512000
  Descriptor Blocks:
  10.0.0.2 (Serial0/0/0), from 10.0.0.2, Send flag is 0x0
      Composite metric is (40512000/28160), Route is Internal
      Vector metric:
        Minimum bandwidth is 1544 Kbit
        Total delay is 21000 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
```

2.29 Lab 54: EIGRP Neighbor Troubleshooting

Objective

Troubleshoot EIGRP neighbor relationships.

Common EIGRP Issues

Issue	Solution
AS number mismatch	Verify AS number matches on all routers
K-values mismatch	Check metric weights (must match)
Authentication failure	Configure same key chain
Network statements missing	Verify networks are included
Split horizon blocking	Disable on hub-and-spoke: `no ip split-horizon eigrp 100`
Passive interface	Check if interface is passive

Troubleshooting Commands

```
! Check neighbor status
R1# show ip eigrp neighbors
! If no neighbors, check:
! 1. Interface status
R1# show ip interface brief

! 2. EIGRP process
R1# show ip protocols

! 3. EIGRP interface settings
R1# show ip eigrp interfaces

! Debug EIGRP
R1# debug eigrp packets
R1# debug eigrp neighbors

! K-value mismatch example
R1# show ip eigrp neighbors
*Mar 15 14:30:15.123: EIGRP: Neighbor 10.0.0.2 not on common network for AS 100
*Mar 15 14:30:15.124: EIGRP: K-values mismatch
! Fix by matching metric weights
R1(config-router)# metric weights 0 1 0 1 0 0
R2(config-router)# metric weights 0 1 0 1 0 0

! Clear EIGRP neighbors
R1# clear ip eigrp neighbors
```

2.30 Lab 55: DHCP Configuration on Router

Objective

Configure router as DHCP server for LAN clients.

Topology

```
[PC1]---[Switch]---[R1]
(DHCP Client)      (DHCP Server)
```

Configuration Steps

```
! Create DHCP pool
R1(config)# ip dhcp pool LAN_POOL
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1
R1(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
R1(dhcp-config)# domain-name ccnadomain.com
R1(dhcp-config)# lease 7 0 0
R1(dhcp-config)# exit

! Exclude static addresses
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
R1(config)# ip dhcp excluded-address 192.168.1.254

! Configure interface for DHCP relay (if needed)
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown

! Verify DHCP
R1# show ip dhcp binding
IP address        Client-ID/              Lease expiration        Type
                  Hardware address
192.168.1.11      0063.6973.636f.0001.    Mar 22 2024 02:30 PM
   Automatic
                  000a.000b.000c--.ab12

R1# show ip dhcp pool
Pool LAN_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 11
 Pending event                  : none
 1 subnet is currently in the pool
 Current index        IP address range                    Leased
   addresses
 192.168.1.11         192.168.1.1        - 192.168.1.254   1

! Test on PC
PC> ipconfig /dhcp
IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1
```

```
DNS Servers: 8.8.8.8, 8.8.4.4
```

2.31 Lab 56: DHCP Relay Agent

Objective

Configure DHCP relay to forward requests across subnets.

Topology

```
[PC1]---[R1]---(Network)---[R2]---[DHCP Server]
(VLAN10)                                   (192.168.100.10)
```

Configuration Steps

```
! On R1 (relay agent)
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# ip helper-address 192.168.100.10
R1(config-if)# exit

! On R2 (router between)
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.100.1 255.255.255.0
R2(config-if)# no shutdown

! On DHCP Server (can be router or actual server)
! Configure DHCP pool for remote subnet
DHCP-Server(config)# ip dhcp pool REMOTE_POOL
DHCP-Server(dhcp-config)# network 192.168.1.0 255.255.255.0
DHCP-Server(dhcp-config)# default-router 192.168.1.1
DHCP-Server(dhcp-config)# exit

! Verify relay
R1# show ip interface gigabitEthernet 0/0 | include helper
Helper address is 192.168.100.10

! Debug DHCP relay
R1# debug ip dhcp server events
R1# debug ip udp 67
```

Ports Used by DHCP

Port	Usage
UDP 67	DHCP server
UDP 68	DHCP client

2.32 Lab 57: DNS Configuration Basics

Objective

Configure DNS on router and test name resolution.

Configuration Steps

```
! Configure DNS server on router
R1(config)# ip domain-lookup
R1(config)# ip name-server 8.8.8.8
R1(config)# ip name-server 8.8.4.4

! Configure static DNS entry
R1(config)# ip host R2 10.0.0.2
R1(config)# ip host webserver 192.168.1.100
R1(config)# ip host mailserver 192.168.1.101

! Verify DNS resolution
R1# ping R2
Translating "R2"...domain server (8.8.8.8) [OK]
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!

! Test DNS lookup
R1# ping webserver
Translating "webserver"...domain server (8.8.8.8) [OK]
Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
.....

! Show DNS cache
R1# show hosts
Default domain is not set
Name/address lookup uses domain service
Name servers are 8.8.8.8, 8.8.4.4

Host                  Flags       Age Type   Address(es)
R2                    (temp) (0)  0    IP     10.0.0.2
webserver             (perm) (0)  0    IP     192.168.1.100
mailserver            (perm) (0)  0    IP     192.168.1.101

! Clear DNS cache
R1# clear host
```

2.33 Lab 58: NTP Configuration

Objective

Configure NTP for accurate time synchronization.

Topology

```
[R1]---(Network)---[NTP Server]
(NTP Client)       (192.168.1.100)
```

Configuration Steps

```
! Configure NTP client
R1(config)# ntp server 192.168.1.100
R1(config)# ntp server 0.pool.ntp.org

! Configure router as NTP server
R1(config)# ntp master 2
! (stratum level, lower is more accurate)

! Authenticate NTP
R1(config)# ntp authenticate
R1(config)# ntp authentication-key 1 md5 cisco123
R1(config)# ntp trusted-key 1
R1(config)# ntp server 192.168.1.100 key 1

! Verify NTP
R1# show ntp status
Clock is synchronized, stratum 3, reference is 192.168.1.100
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is
    2**10
reference time is DF1F7B5C.12345678 (14:30:15.123 EST Thu Mar 15 2024)
clock offset is 0.1234 msec, root delay is 12.34 msec
root dispersion is 23.45 msec, peer dispersion is 5.67 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is
    0.0000123456 s/s

! Show NTP associations
R1# show ntp associations
  address         ref clock     st   when   poll   reach   delay
    offset   disp
*~192.168.1.100   .GPS.          1     23     64    377    12.34
    0.12    5.67
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~
    configured

! Set timezone
R1(config)# clock timezone EST -5
R1(config)# clock summer-time EDT recurring
```

2.34 Lab 59: Syslog Configuration

Objective

Configure syslog to send logs to central server.

Configuration Steps

```
! Configure syslog server
R1(config)# logging host 192.168.1.100
R1(config)# logging trap informational
R1(config)# logging source-interface loopback 0

! Logging levels (0-7)
! 0 - Emergency (system unusable)
! 1 - Alert (immediate action)
! 2 - Critical (critical condition)
! 3 - Error (error condition)
! 4 - Warning (warning condition)
! 5 - Notification (normal but significant)
! 6 - Informational (informational messages)
! 7 - Debugging (debug messages)

! Configure logging options
R1(config)# logging on
R1(config)# service timestamps log datetime msec localtime
R1(config)# logging buffered 16384 informational
R1(config)# logging console warnings
R1(config)# logging monitor debugging

! Verify logging
R1# show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0
    flushes, 0 overruns)
    Console logging: level warnings (4)
    Monitor logging: level debugging (7)
    Buffer logging: level informational (6), 16384 bytes logged
    Trap logging: level informational (6), 32 message lines logged
        Logging to 192.168.1.100 (udp port 514, audit disabled,
             authentication disabled, encryption disabled)

! Generate test log
R1# send log "Test syslog message"
```

2.35 Lab 60: SNMP Basics

Objective

Configure SNMP for network monitoring.

Configuration Steps

```
! Configure SNMP community strings
R1(config)# snmp-server community public RO
R1(config)# snmp-server community private RW

! Set SNMP location and contact
R1(config)# snmp-server location "Data Center - Rack A"
R1(config)# snmp-server contact "admin@ccnadomain.com"

! Configure SNMP traps
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 192.168.1.100 version 2c public

! SNMP versions
! v1 - Original, least secure
! v2c - Uses community strings (public/private)
! v3 - Authentication and encryption

! SNMPv3 configuration
R1(config)# snmp-server group ADMIN v3 priv
R1(config)# snmp-server user admin ADMIN v3 auth md5 cisco123 priv des
    cisco123

! Verify SNMP
R1# show snmp community
Community name: public
Community name: private

R1# show snmp group
groupname: ADMIN                              security model:v3 priv
contextname: <no context specified>           storage-type: permanent
readview : <no readview specified>            writeview: <no writeview
    specified>
notifyview: <no notifyview specified>
row status: active

R1# show snmp user
User name: admin
Engine ID: 800000090300001122334455
storage-type: permanent
Authentication Protocol: MD5
Privacy Protocol: DES
Group-name: ADMIN
```

Appendix

A: Cisco IOS Command Hierarchy

```
User EXEC Mode (>)           - Limited monitoring
    | enable
Privileged EXEC Mode (#)     - Full access to device
    | configure terminal
Global Configuration Mode    - Device-wide settings
    | interface <name>
Interface Configuration Mode - Port-specific settings
    | router <protocol>
Routing Protocol Mode        - Protocol-specific settings
    | line <type> <range>
Line Configuration Mode      - Console/VTY settings
```

B: Common Port Numbers

Port	Service
TCP/20,21	FTP
TCP/22	SSH
TCP/23	Telnet
TCP/25	SMTP
UDP/53	DNS
UDP/67,68	DHCP
UDP/69	TFTP
TCP/80	HTTP
TCP/110	POP3
UDP/123	NTP
TCP/443	HTTPS
UDP/514	Syslog
UDP/161,162	SNMP

C: Subnetting Cheat Sheet

Prefix	Subnet Mask	Wildcard Mask
/8	255.0.0.0	0.255.255.255
/16	255.255.0.0	0.0.255.255
/24	255.255.255.0	0.0.0.255
/25	255.255.255.128	0.0.0.127
/26	255.255.255.192	0.0.0.63
/27	255.255.255.224	0.0.0.31
/28	255.255.255.240	0.0.0.15
/29	255.255.255.248	0.0.0.7
/30	255.255.255.252	0.0.0.3
/31	255.255.255.254	0.0.0.1
/32	255.255.255.255	0.0.0.0

D: Useful Shortcuts

```
Ctrl+Shift+6 (then X) - Suspend Telnet/SSH session
Ctrl+Z - Exit config mode to privileged EXEC
Ctrl+C - Abort command
Ctrl+R - Redisplay line
Tab - Complete command
? - Context-sensitive help
```

E: Lab Completion Checklist

Use this checklist to track your progress through all 100 labs:

Level	**Labs Completed**
Beginner (1-25)	__/25
Intermediate - Switching (26-41)	__/16
Intermediate - Routing (42-54)	__/13
Intermediate - Services (55-60)	__/6
Advanced - NAT & ACL (61-69)	__/9
Advanced - WAN & Routing (70-79)	__/10
Advanced - Security (80-86)	__/7
Advanced - Services (87-91)	__/5
Advanced - Troubleshooting (92-100)	__/9

F: Recommended Resources

- Cisco Packet Tracer (Free for NetAcad students)
- GNS3 (Advanced, requires IOS images)
- EVE-NG (Professional virtual lab)
- Cisco Modeling Labs (CML) - Paid
- Boson NetSim (Exam simulation)

Conclusion

Congratulations on completing all 100 CCNA labs! You have now mastered:

- Basic device configuration and management
- Switching technologies (VLANs, STP, EtherChannel)
- Routing protocols (Static, RIP, OSPF, EIGRP)
- Network services (DHCP, DNS, NTP, Syslog)
- Security features (SSH, ACLs, Port Security)
- WAN technologies (PPP, GRE)
- Troubleshooting methodologies

Next Steps:

1. Review weak areas using your lab notes
2. Take practice exams (Boson, MeasureUp)
3. Build a full enterprise lab combining all technologies
4. Schedule your CCNA 200-301 exam

Good luck on your CCNA journey!